Northstar Components

Next-Period Production Plan — Management Report

Warehouse NSC-01 · Currency EUR · Prepared 18 July 2026 · Single-solver optimization task

Executive Summary

Northstar Components must set production quantities for two products — A (Precision housing) and B (Control module) — for the coming period so as to maximize total contribution margin without breaching any hard resource capacity, product minimum commitment, or approved demand cap.

The recommended plan is to produce 24 units of Product A and 3 units of Product B, yielding a maximum total contribution of EUR 1,125. This solution was computed with the Optees mixed-integer linear programming solver (milp.linear, contract v1), which reported a mathematically optimal status, and it was independently verified as feasible and objective-consistent by the Optees validator.

Machine time is the single binding resource: the plan consumes all 60 available machine hours. Product A is produced at its full approved demand cap of 24 units, while Product B is held at its mandatory minimum commitment of 3 units. Raw material and direct labour retain slack, meaning additional machine capacity — not material or labour — is the lever that would unlock further contribution.

Recommended Production Quantities

Product	Description	Recommended units	Unit contribution (EUR)	Contribution (EUR)
A	Precision housing	24	40	960
B	Control module	3	55	165
Total		27		1,125

All quantities are whole units, satisfying the workbook requirement that production be expressed in integer units.

Expected Contribution

Total expected contribution for the period is EUR 1,125, composed of EUR 960 from Product A (24 units × EUR 40) and EUR 165 from Product B (3 units × EUR 55). Contribution is defined in the workbook data dictionary as revenue less variable cost per produced unit; fixed costs are outside the scope of this optimization.

Resource Utilization

Resource	Used	Available	Utilization	Status
Machine time (h)	60	60	100%	Binding
Raw material (kg)	78	80	97.5%	Slack: 2 kg
Direct labour (h)	30	35	85.7%	Slack: 5 h

Usage is derived from the per-unit coefficients in the Products sheet: machine 2A+4B, material 3A+2B, labour 1A+2B, evaluated at A=24, B=3.

Binding Constraints

The optimal plan is limited by the following active constraints:

- Machine time capacity (60 hours) is fully consumed and is the sole binding resource constraint. It is the effective bottleneck of the plan.
- Product A demand cap (24 units, approved) is binding — Product A is produced at its maximum sellable quantity.
- Product B minimum commitment (3 units) is binding — Product B is held at its mandatory floor because each additional B unit consumes 4 machine hours (versus 2 for A) while machine time is already exhausted.

Raw material and direct labour are non-binding and retain slack of 2 kg and 5 hours respectively.

Assumptions

- Demand limits are taken from the Direct Demand Plan sheet (A ≤ 24, B ≤ 10), both marked Approved, as instructed for the single-solver task. Historical-demand sheets (History A, History B, Next Period) were not used.
- Minimum production commitments (A ≥ 5, B ≥ 3) are treated as hard lower bounds per the Products sheet and data dictionary.
- Hard resource capacities (machine 60 h, material 80 kg, labour 35 h) are treated as hard ≤ constraints, consistent with the Resources sheet flag 'Hard constraint = True'.
- Unit contributions (A = EUR 40, B = EUR 55) and per-unit resource coefficients are taken directly from the Products sheet.
- Production quantities are integer-valued (whole units required).
- Administrative Notes fields (warehouse code, currency, colour, energy review) are contextual only and were not converted into constraints, per the workbook's explicit guidance.

Limitations

- The model is a static single-period plan; it does not consider inventory carryover, setup times, or multi-period dynamics.
- The 'Energy review scheduled next quarter' note supplies no numeric limit, so no energy constraint was modelled. Should an energy cap be issued, the plan must be re-optimized.
- Data are fully synthetic; the figures are illustrative and not drawn from a real company.
- The Optees validator confirms feasibility, integrality, and objective consistency, but notes that these checks do not by themselves prove optimality, nor do they assess whether the model faithfully represents business intent. Optimality here rests on the solver's own status.

Optees Validation Status

Item	Result
Capability	milp.linear (contract v1, problem/result schema v1)
Payload validation	valid — no warnings
Solver backend	Google CP-SAT (via optees.milp_router)
Mathematical status	OPTIMAL (relative gap 0)
Reported objective	EUR 1,125 (recomputed 1,125; difference 0)

Item	Result
Independent validation	VERIFIED — all 5 checks passed, 0 violations
Checks passed	variable vector, bounds, integrality, constraints, objective

Mathematical status (solver-reported optimality) and independent validation (feasibility and objective consistency) are reported separately above. The solver reports the solution as provably optimal; the validator independently confirms the solution is feasible and that the stated objective matches the recomputed objective.